Verb doubling and the architecture of realization*

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Colloquium, Yale University, April 25 2022

1 Introduction

Minimalism proposes that syntax is constrained by conditions at the **interfaces**.

- Open question whether interfaces are merely points of transition, or themselves internally complex.

Competing—and often incommensurate—models of how form maps onto meaning.

In this talk:

- Empirical focus: verb doubling motivated by prosodic needs of a clitic / particle.
  - Examples from Breton, Hakka, Ingush, and Malayalam
- Using this to make several claims about the architecture of PF:
  1. Linearization is **constraint based**, and proceeds in parallel with prosodification
  2. The interaction of constraints can trigger some instances of **doubling**, in order to provide clitics with hosts.
  3. This doubling crucially occurs **prior** to Vocabulary Insertion (which is in turn prior to segmental phonology)

Outline for the talk:

§2: Theoretical context

§3: Prosodically motivated verb doubling: the facts

§4: Analysis: Prosodically motivated verb doubling in staged realization

§5: Discussion and Extensions

§6: Conclusion

*Thanks are due to many people for insightful questions and discussion, in particular Neil Banerjee, Tanya Bondarenko, Emily Clem, Norvin Richards, Nicholas Rolle, Rachel Walker, and Danfeng Wu. Thanks also to audiences who have given feedback on earlier versions of this work, at the Workshop on Morphological Repairs at the University of Leipzig in 2018, at NELS 50 in 2019, at QMUL, University of Tromsø, UMass Amherst, and UCSC in early 2020, and most recently at MIT in 2022. Thanks finally to Dan Siddiqi for ongoing insightful discussion about the mechanics and metatheory of morphological realization.

2 Theoretical context: what is it to be "at PF"?

In GB/P&P: sub-systems in syntax, including language-specific operations or principles.

Minimalist turn: small number of invariant mechanisms in syntax, and attributed where possible to conditions imposed by interfaces with meaning (LF) and form (PF)—or to "third factors" in acquisition.

- variation attributed to differences in the lexicon / set of formal features active in a language (the Borer conjecture, Borer 1984)

Subsequent developments: **variation within the interfaces**—at least at spell-out / PF.

- Syntactic operations moved "to PF": head movement, case valuation, agreement, etc.

  → uncharitably: pushing inconvenient operations to PF in order to ignore them.

  → more charitably: language-specific rules/repairs, linearization, prosody, etc.

Distributed Morphology, as a theory, adds further operational machinery post-spellout.

→ Impoverishment, Fusion, Fission, Local Dislocation, Vocabulary Insertion, readjustment rules, etc.—all of which are expected to apply differently in different languages.

If the PF interface includes phonology, potential for more internal structure (levels/cycles).

This "interface" has gotten very large indeed.

Questions worth asking:

- What do we gain by moving operations to PF?
- When things happen "at PF", do they happen in sequence or in parallel? If sequential, in what sequence?
- In this kind of framework, can we be Minimalist about the content of PF itself?

PF: clearing out the junk drawer

Constraining the space of possibility: some starting assumptions

- "Narrow syntax" operates over formal features and roots; these are associated with phonological content by a post-syntactic process of Vocabulary Insertion (Distributed Morphology, DM: Halle and Marantz 1993, 1994).

  Syntactic representations encode hierarchical constituency, but not linear order.

  (Principle of phonology-free syntax: Zwicky 1969)

  (cf. Richards 2010, 2016, where the syntax knows a bit about phonology)

- "Narrow phonology" operates over linear representations, and lacks access to hierarchical relations such as relative scope or headedness

  (Bjorkman and Dunbar, 2016; Jurgec and Bjorkman, 2018).

  Phonological inputs include prosodic constituents that correspond (potentially imperfectly) to syntactic constituents, but not morphosyntactic trees.

  (Aside: is "narrow phonology" **on** the PF branch or **after** it?)

What is minimally conceptually necessary for PF?

- An algorithm that takes a tree as input and produces a string as output.

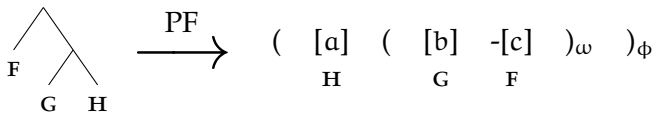


Figure 1: PF: a conversion between trees and strings *(to be revised)*

Conflict among existing proposals:

- Linearization is / is not constraint-based
  - Deterministic algorithm (Kayne 1994; Fox and Pesetsky 2005; Citko 2005, 2011 a.o.)
  - Linearizing algorithm is a violable constraint: e.g. Optimal Linearization (Kusmer, 2020, 2019)
- Vocabulary Insertion is / is not cyclic
  - Fully cyclic, bottom-up: Bobaljik (2000); Embick (2010); Myler (2017); Kalin (2021)
  - Non-cyclic (global constraint evaluation): Distributed Optimality (Trommer, 2002); OT-DM (Rolle, 2020); Optimal Interleaving (Wolf, 2008)
  - Non-cyclic (post-linearization): Haugen and Siddiqi (2016), Ostrove (2018)
- Narrow phonology does / does not have direct access to syntax
  - Does, via cyclicity (constrained): Phase-based phonology (Wagner, 2005; Newell, 2008)
  - Does, via strict prosody-syntax correspondence: Direct reference theories (e.g. Wagner 2005)
  - Imperfectly, via violable prosody-syntax correspondence: Match Theory (Selkirk, 2011; Elfner, 2012, 2015)

Charting a path through these debates: staged spell-out at PF

1. Linearization + prosodic mapping

   INPUT: Output of narrow syntax: hierarchy of heads consisting of features and roots
   OUTPUT: String of terminals (bearing features) with prosodic boundaries

   Replacive: hierarchical information is absent from the output.

   Constraint-based: candidates evaluated according to ranked and violable constraints, as in OT.

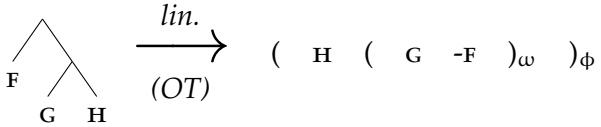


Figure 2: Linearization + prosodic mapping: constraint-based evaluation

   RESPONSIBLE FOR: (some) affix / clitic / particle distinctions; local dislocation; fusion (perhaps impoverishment and fission)

2. Vocabulary Insertion

   INPUT: String of terminals (bearing features)
   OUTPUT: Input to narrow phonology: string of terminals associated with phonological content (including empty syllables/moras, floating tones, etc.)

   Additive: features and prosodic boundaries are not lost, but phonological content is added to them.

   Rule-based: VI takes place sequentially ("inside out"), via realization rules (themselves ordered)

$$(\ \textsc{h}\ (\ \textsc{g}\ \ \text{-}\textsc{f}\ )_{\omega}\ )_{\phi} \xrightarrow[\textit{(deriv.)}]{\textit{VI}} (\ \underset{\textsc{h}}{[a]}\ (\ \underset{\textsc{g}}{[b]}\ \underset{\textsc{f}}{\text{-}[c]}\ )_{\omega}\ )_{\phi}$$

Figure 3: Vocabulary Insertion: rule-based derivation

   RESPONSIBLE FOR: Vocabulary Insertion; (perhaps impoverishment and fission)

3. Narrow Phonology

Evidence that VI is prior to (not parallel with) narrow phonology

- Suppletion of infixes can be determined by phonological context at the edge of what they attach to, but not by the phonological context of their surface position (when those differ) (Kalin and Rolle, to appear; Kalin, 2021).
- Evidence that even phonologically-determined allomorphy can be non-optimizing, e.g. **?**

Evidence that VI is after (not parallel with) constraint-based linearization+prosody

- *This talk:* prosodically motivated verb doubling

  "Language-specific repairs" attributed instead to variably ranked and interacting constraints.

  - Contra Kusmer (2019, 2020), for whom linearization must either be after or parallel to VI; also Wolf (2008) where VI is parallel with phonology more broadly.
  - Similarities with Rolle (2020), but dislocation operations attributed to general constraints on linearization, rather than to morphological wellformedness.
- cf. Ostrove (2018): VI based on linearized string.

  (*Not addressed today:* linear alternatives to cyclic VI)

3 Prosodically motivated verb doubling

The past 20 years has seen growing attention to verb doubling in predicate fronting contexts:

- Nunes (1999, 2004): Copy Theory of movement requires that we rule out doubling—can predict doubling when it does not conflict with linearization.
  - Much work following in this tradition: Landau (2006); Kandybowicz (2007, 2008); Kandybowicz and Torrence (2015); Aboh and Dyakonova (2009); Trinh (2011); Hein (2017, 2020); Kandybowicz and Torrence (2021)

Predicate fronting variously involves movement of V or VP to a topic or focus position:

(1) Hebrew: fronting of V or VP (Landau, 2006)

a. [li-**knot** et ha-praxim], hi **kanta**.
[INF-buy ACC the-flowers] she buy.PAST
"As for buying the flowers, she bought."

b. [li-**knot**], hi **kanta** et ha-praxim.
[INF-buy] she buy.PAST ACC the-flowers
"As for buying, she bought the flowers."

Common analysis: something about predicate fronting allows multiple copies to be realized

- Tension between two strands of approach:
  - Something about the movement chain breaks the link between two copies, so that they aren't treated as such for linearization.
  - Doubling arises as a trade-off between constraints preferring single-copy realization and constraints on other aspects of well-formedness

Today: Cases of verb doubling that have a more "last resort" character: not the result of movement, but of other factors.

These can be captured straightforwardly by the interaction of constraints governing the position of prosodically weak elements (clitics / particles) and those responsible for **prosodically-motivated reduplication**—but that this doubling crucially occurs **prior** to Vocabulary Insertion.

The phenomenon of interest: doubling motivated by **prosodic needs of a clitic/particle**.

- Trigger for doubling is prosodic → **post-syntactic**
- Copies can be non-identical → **not segmental copying, pre-Vocabulary Insertion**

First: illustrations from Hakka and Malayalam.

Then: more detailed discussion of Breton and Ingush.

3.1 Hakka: doubling with adjunct CP

Hakka (Sino-Tibetan; Southern China, Taiwain)

Verb doubling triggered by an enclitic complementizer *do* (Tseng 2008)

Lai 2002: *do* is an enclitic complementizer that introduces a manner or result clause, grammaticalized from a verb meaning *be at* or *arrive*.

Tseng (2008): when *do* is not adjacent to the main verb (because an object or PP intervenes), the verb doubles.[1]

(2) a. Gi sii [do dong kiak].
he eat COMP really fast
"He ate really fast."

[1] Hakka has 6 grammatical tones (Hashimoto, 1973), but tone is not shown in the examples here drawn from Tseng (2008).

b. Gi sii shui-go **sii** [do dong kiak].
he eat fruit eat COMP really fast
"He ate fruit really fast."
c. *Gi sii sii [do dong kiak]. (Tseng 2008, 53–54)

Compound verbs with no object optionally double one part of the compound before *do*; full doubling is ungrammatical.

(3) a. Gi fun-mi (**fun**) [do tai-ga dong dam-sim].
he in.a.coma COMP everyone really worried
"He's in a coma, which makes everyone worry about him."
b. *Gi fun-mi **fun-mi** [do tai-ga dong dam-sim]. (Tseng 2008, 67–68)

By contrast, when a compound verb is not adjacent to *do*, doubling is obligatory—but can be either partial or total.

(4) a. Gi fun-mi [di yi-yen] **fun(-mi)** [do tai-ga dong dam-sim].
he in.a.coma [PREP hospital] in.a.coma COMP everyone really worried
"He's lying in the hospital with a coma, which makes everyone so worried."
b. *Gi fun-mi [di yi-yen] [do tai-ga dong dam-sim]. (Tseng 2008, 68)

Hashimoto (1973, 495) describes similar doubling apparently triggered by a verbal suffix *-det*^{5}, which appears to introduce degree adverbials (*an*^{3} *kvai*^{4} "so fast", *ho*^{3} *ho*^{3}, "very well")

3.2 Malayalam:

Malayalam (Dravidian; Southern India)

Verb doubling triggered by emphatic particle *-ee*, alternating with DO-support.

(all Malayalam examples provided by Athulya Aravind, any errors my own)

- Suffix *-ee* is an NPI, meaning something like "at all"; other inflectional information can be expressed in one of three ways: as an independent word; on a support verb *ceyy* ("do"); on a second copy of the main verb.

(5) Raman chooru thinn-illa
R. rice eat-NEG
"Raman didn't eat rice."

(6) Raman chooru thinn-ee illa
R. rice eat-EMPH NEG
"Raman didn't eat (any) rice at all."
(negation as separate word)

(7) Raman chooru thinn-ee cheyt-illa
R. rice eat-EMPH do-NEG
"Raman didn't eat (any) rice at all."
(DO-support)

(8) Raman chooru **thinn**-ee thinn-illa
R. rice eat-EMPH eat-NEG
"Raman didn't eat (any) rice at all."
(**doubling**)

(9) Anita pustakam vaayich-oo?
A. book read.PFV-Q
"Did Anita read the book?"

(10) Anita pustakam vaayich-ee cheyt-oo?
A. book read.PFV-EMPH do-Q
"Did Anita read the book at all?"
(neg. bias; DO-support)

(11) Anita pustakam **vaayich**-ee vaayich-oo?
A. book read.PFV-EMPH read-Q
"Did Anita read the book at all?"
(neg. bias; **doubling**)

3.3 Ingush: doubling at the *v*P edge

Ingush (Nakh-Dagestanian; Caucasus)

Of interest today: doubling triggered by the preverbal enclitic =ʔ*a*

(Peterson 2001; Conathan and Good 2000; Nichols 2011)

3.3.1 =ʔ*a* with and without verb doubling

Properties of =ʔ*a*:

- occurs in **clause chaining**; following Nichols (2011) labelled here &

  (=ʔ*a* also used for emphasis, nominal focus, but without doubling)

- requires a **vP-internal** host: direct object, adverb, deictic clitic, first element in light / compound verb.
- If no such element is present, the verb **doubles** to provide a host.

  (doubling also attested with emphasis / verum focus interpretation)

Note: Same pattern found in related Chechen (Conathan and Good, 2000); Avar, Andic (Nichols and Peterson, 2010)

(12) =ʔ*a* without doubling: other host available

a. [Baga hwalt'am =ʔa dellaa] vaagha.
[mouth.ADV dumpling=& AGR.insert.ACV] AGR.sit.PRES
"He's sitting there with a dumpling in his mouth." (Nichols 2011, 373)

b. Muusaa [gaalie-ča banaana ʕa=či =ʔa jillaa] vaxar.
Musa [bag-LOC banana down=in=& AGR.put.ACV] AGR.go.PAST
"Musa put the banana in the bag and left." (Peterson 2001, 148)

c. muusaa [k'eank ciec =ʔa veaqqaa] v-ax-ar.
Musa [boy surprise=& AGR.AUX.ACV] AGR.go.PAST
"Musa surprised the boy and left." (Peterson 2001, 152)

When no other host is available, the **verb doubles** to host =ʔa:

(13) a. Object of chained verb hosts =ʔa, no doubling
Muusaa [buc =ʔa hʕaq-aa] či=v-ie-r.
Musa [grass=& mow-ACV] in=AGR-go-PAST
"Musa cut the grass and went home."

b. Object shared with matrix verb (subject=ERG), verb doubles to host =ʔa
Muusaa-z buc [hʕaq =ʔa hhaq-aa] ci=j-ie-r.
Musa-ERG grass **mow**=& mow-ACV in=AGR-go-PAST
'Musa cut the grass and brought it home. (Peterson, 2001, 149)

(14) jett [**laq** =ʔa **laq**-aa] b-el-ar.
cow [go.dry=& go.dry-ACV] AGR-die-PAST
"The cow stopped giving milk and died." (Peterson 2001, 147)

(15) je kinashjka diisha='a diishaa
this book D(AGR).read=& D(AGR).read.ACV
"they (SG) read this book and..." (Nichols 2011: 219)

- The **stem** of the main verb is copied, without inflectional suffixes.
- Yet the copies **need not be segmentally identical**:

(16) Muusaa, balkha **ga** =ʔa **gejn**, avtovusaa t'eħa-vysar
Musa work.ADV delay=& delay.ACV bus miss.PAST
"Musa was hung up at work and missed the bus." (Conathan and Good 2000, 53)

- Conathan and Good (2000): for irregular verbs including *gaa* 'delay' in (16), *gu* 'see', and *lie* 'die', the double that supports =ʔa is non-identical to the inflected verb.
- They give no examples for *gu* or *lie*, but Nichols (2011) gives examples of focus-induced verb doubling (triggered by negation) that show the relevant stem suppletion for *lie* 'die':

(17) **Vala** cy **lezh** vyssaav yz?
die NEG die.SCV remain.PAST(non-witnessed) 3SG
"He didn't die?" (*lit.* "He remained not dying?")
(Listener double-checks what narrator said.) (Nichols 2011: 223)

3.3.2 The domain of =ʔ*a*:

- **Dative** objects cannot host =ʔa.

(18) a. muusaa k'eank-ea siirda =ʔa siird-aa, v-ax-ar.
Musa boy-DAT swear.at =& swear.at-ACV, AGR-leave-PAST
"Musa swore at the boy and left." (Peterson 2001, 149)
b. *muusaa k'eank-ea =ʔa siird-aa, v-ax-ar.
Musa boy-DAT =& swear.at-ACV, AGR-leave-PAST
"Musa swore at the boy and left." (Peterson 2001, 150)

- Assuming dative objects introduced by ApplP → domain of attachment is smaller constituent, **vP**

  →clause chaining as vP coordination

3.3.3 Previous analyses of Ingush verb doubling

Peterson (2001): =ʔ*a* within a typology of clitics

- typologically unusual second-from-end enclitic: Type 5 Klavans (1985)

Conathan and Good (2000): Construction-based approach within HPSG

- Both linear order and duplication defined directly in the syntax:
  - =ʔa occurs in a fixed pre-verbal position (second-to-last)
  - One of the frames of =ʔa is as the second element a VP co-headed by two verbs
  - Morphological features are specified directly in the syntax: first verb is infinitive, second is finite (and agrees appropriately)
- Analysis does not capture the intuition that doubling is *triggered* by the clitic

3.4 Breton: doubling at the clause edge (V2)

Breton (Celtic; Brittany)

Of interest today: second position *rannig* element. (Jouitteau 2005, 2011, 2012)

3.4.1 The *rannig* with and without verb doubling

- "Linear" V2 requirement in main finite clauses (Borsley and Kathol, 2000)
- V2 requirement imposed by *rannig*, which heads FinP (Jouitteau, 2005)

  Satisfied by elements in a variety of structural positions: focused phrase (in Spec-FocP); topic (in Spec-TopP); negation (NegP); non-finite main verb (in presence of auxiliaries).

(19) a. [Ar vugale] o deus gwalc'het ar wetur dec'h.
[the children] R.3PL have wash.PTCP the car yesterday
"The children washed the car yesterday."
b. [Gwalc'hin ar wetur] o deus graet.
[wash.INF the car] R.3PL have do.PTCP
"They really did wash the car."
c. [Dec'h] o deus ar vugale gwalc'het ar wetur.
[yesterday] R.3PL have the children wash.PTCP the car
"The children washed the car yesterday." (Schafer 1994, 24)

(20) [D' ar jardin] ez an.
[P DET garden] R go.1SG.
"I am going into the garden." (Jouitteau 2012)

- In the absence of another clause-initial element, (some) verbs can **double** to provide a pre-*rannig* element.

(21) a. **Mont** a **yan** d' ar jardin.
go R go.1SG P DET garden
"I am going into the garden."
b. **Redek** a **redan** bemdez.
run.INF R run.1SG every.day
"I run every day."
c. **Dleout** a **zlean** ober ma gwele.
must.INF R must.1SG do my bed
"I have to make my bed." (Jouitteau 2012)

- Again, leftmost copy of the verb is **non-finite** (though in Breton a special form)
- And two copies can show different suppletive forms

However:

- Doubling in Breton is **restricted to a small set of verbs**:

  Differences across varieties, but including: *ober* 'do'; *bezañ* 'be'; *rankout* and *dleout* 'must'; *gallout* 'can', *dont* 'come'; *mont* 'go', *gouzout* 'know'; *kerzhout* 'walk', *redek* 'run', and *lenn* 'read'

Doubling contrasts with an "analytic construction" with *ober* 'do':

(22) a. **Mont** [a] **yan** d' ar jardin.
go R go.1SG P DET garden
"I am going into the garden."
b. **Mont** [a] **ran** d' ar jardin.
go R do.1SG P DET garden
"I am going into the garden."

Unlike doubling, analytic form in 'do' is fully productive (except with *bezañ* 'be' and *kaout* 'have').

- When both are available, doubling is information-structure neutral; DO-support associated with predicate focus.
- When only DO-support is available, it can have both neutral and focused interpretations.

3.4.2 The *rannig* as a clitic

Unlike in Ingush, the *rannig* is not obviously enclitic—but also not obviously proclitic to the verb:

- With *kaout* 'have', pre-verbal clitics (~ subject agreement) fuse with *rannig*.

(19-b) [Gwalc'hin ar wetur] [o] deus graet.
[wash.INF the car] R.3PL have do.PTCP
"They really did wash the car."

- However, pre-verbal object clitics (proclitic to verb) do not fuse with *rannig*. If *rannig* is a proclitic, we might expect it to fuse with object clitics when adjacent, like it does in (22).

However: *rannig* is an *unaccented monosyllable* that cannot occur clause-initially.

Prosodically weak element; adopting the label "clitic" to describe this distribution

3.4.3 Previous analysis of Breton verb doubling

Jouitteau (2011, 2012)

- Main goal is to account for analytic form with 'do'
- Assumes that V moves to Fin, forming complex head: R(Obj.Cl)-V-T
- Both 'do' and doubling reflect **post-syntactic excorporation**:
  - Finiteness head cannot be clause initial.
  - As a last-resort mechanism the verb can **excorporate** to provide a pre-*rannig* element.

Questions for this analysis:

- *Why* can the finite verb not be clause initial?
- What is the nature of "excorporation"?

3.5 Summary

In all these languages, a functional particle can trigger **doubling**:

Doubling is a "last resort" repair in the absence of another potential host.

Points to note:

- Doubling has no semantic effect when it is obligatory, and is not freely available
- Non-identity of the two stems in Ingush and Breton excludes purely segmental reduplication—despite the motivation for doubling being phonological.

4 Analysis: prosodically-motivated doubling in staged PF realization

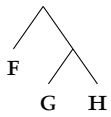
Reduplication often occurs as a **prosodic repair**.

- To meet minimal word requirements, provide syllable onsets, etc. (McCarthy and Prince, 1993, Yu, 2005, Saba Kirchner, 2010, a.o.)

Prosodically-motivated doubling is reminiscent of this kind of reduplication. However:

- Two copies are not linearly adjacent
- Two copies need not be segmentally identical

The staged architecture at PF:



output of syntax (potentially phase-by-phase)



Linearization + Prosodification

(H (G -F)$_\omega$)$_\phi$ *intermediate linear representation*



Vocabulary Insertion

([a] ([b] -[c])$_\omega$)$_\phi$ *input to segmental phonology*
H G F

- Vocabulary Insertion ideally the sole operation of a distinct morphological computation (cf. Haugen and Siddiqi 2016)
- Intermediate linear representation minimally necessary to encapsulate operations; different levels operate over different elements and produce different outputs.
  - → linearization over syntactic atoms + hierarchy
  - → segmental phonology over segments + strings.

  (contrast Stratal OT, Kiparsky 2000, 2007; Bermúdez-Otero 1999)
- Doubling with suppletion requires that doubling takes place **prior** to VI
- Sensitivity to prosodic well-formedness requires it be no earlier than prosodification.
  - → Parallel with linearization + prosodification

How does linearization + prosodification know about clitics pre-VI?

- Match Theory (Selkirk, 2011; Elfner, 2012, 2015): MATCH constraints require that the terminals within syntactic constituents (heads, phrases, utterances) also be dominated by a single prosodic constituent

  X^0 = word ω XP = prosodic phrase ϕ utterance = intonational phrase ι

- Adaptation: language-specific mapping of categories to prosodic constituents.
  - Proposed independently by Windsor (2017), Guekguezian (2017); see also general proposal by Compton and Pittman (2010) that prosodic words in Inuktitut correspond to DP and CP phases, rather than to heads (with further investigation in Arnhold et al. 2018)

- Functional heads not mapped to phonological words → clitic behaviour (potentially category specific)

  *STRUCTURE ≫MATCH(X_{fn}, ω)

4.1 Analysis: constraint-driven doubling in Ingush

Relevant constraints:

- PROSODIC SUPPORT: A clitic cannot be the first unit in a prosodic phrase.

  (reframed from Franks, 2000)

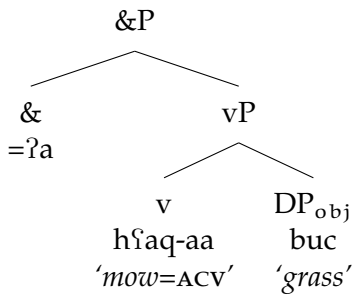
- CLITIC-V: Shorthand for constraints governing position of clitic and V (to be specified)

- UNIFORMITY: No element of S_2 has multiple correspondents in S_1.
  (McCarthy and Prince 1999)

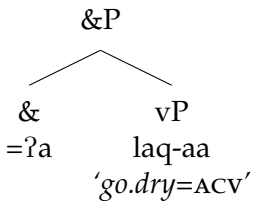
Ranking that induces doubling:

PROSODIC SUPPORT, CLITIC-V ≫ UNIFORMITY

(13-a) Muusaa [buc =ʔa hʕaq-aa] či=v-ie-r.
Musa [grass=& mow-ACV] in=AGR-go-PAST
"Musa cut the grass and went home." (Peterson 2001, 149)



(14) jett [**laq** =ʔa **laq**-aa] b-el-ar.
cow [go.dry=& go.dry-ACV] AGR-die-PAST
"The cow stopped giving milk and died." (Peterson 2001, 147)



What are the constraints governing the position of =ʔa and V in Ingush?

- =ʔa has uses beyond chaining—where there's no doubling:

(23) ʕ**ajšiet j-iilx-ača** =ʔa muusaa v-ax-anz-ar.
Aisha AGR-cry-TCV =& Musa AGR-go-NEG-PAST
"Even when Aisha cried, Musa didn't go." (Concessive)

(24) aaz **qa** =ʔa b-oaqq hʕuoga.
1SG.ERG news =& AGR-communicate 2SG.ALL
"And now I'll tell you some NEWS." (Contrastive focus)

- Order of Clitic-V not required by =ʔa, but by **verb**: final in *v*P prosodic phrase

(25) HEADFINALITY(*v*P) (prosodic implementation; cf. Kusmer 2020, 2019)
Assign a violation if the minimal ϕ-phrase containing *v* in the output is not right-aligned with the minimal ω containing *v*.

- Infixation of =ʔ*a* forced by *STRUCTURE ≫MATCH(X_{fn}, ω), plus wellformedness constraints preventing &P from being mapped to ϕ without containing any prosodic word.

(26) Linearization of (13-a): =ʔa hosted by DP object

[&P =& [vP $\sqrt{\text{mow}}$-*v* DP]]	PR.SUPP.	HEADFINAL(*v*P)	UNIFORMITY
☞ a. (DP=& + $\sqrt{\text{mow}}$-*v*)$_\varphi$			
b. (DP + $\sqrt{\text{mow}}$-*v*=&)$_\varphi$		*!	
c. (DP + $\sqrt{\text{mow}}$=& + $\sqrt{\text{mow}}$-*v*)$_\varphi$			*!

(27) Linearization of (14): no other potential host, verb doubling best option

[&P =& [vP $\sqrt{\text{go.dry}}$-*v*]]	PR.SUPP.	HEADFINAL(*v*P)	UNIFORMITY
a. (=& + $\sqrt{\text{go.dry}}$-*v*)$_\varphi$	*!		
b. ($\sqrt{\text{mow}}$-*v*=&)$_\varphi$		*!	
☞ c. ($\sqrt{\text{go.dry}}$=& + $\sqrt{\text{go.dry}}$-*v*)$_\varphi$			*

After syntax is converted to a linear string with prosodic boundary, handed off to VI:

- Each instance of the root in (27) subject to separate realization.
- Context for stem suppletion met in one but not the other.

4.2 Analysis: constraint-driven doubling in Breton

Relevant constraints:

- PROSODIC SUPPORT: A clitic cannot be the first unit in a prosodic phrase.
- ANTISYMMETRY: Constraint enforcing mapping from syntax to head-initial word order (Kusmer, 2020)

  Assign one violation for each pair of terminal nodes X^0 and Y^0 where:
  a. X^0 asymmetrically c-commands Y^0 and
  b. x does not precede y. (**revised from Kusmer 2020**)

- UNIFORMITY: No element of S_2 has multiple correspondents in S_1.

Evaluating ANTISYMMETRY:

- Kusmer (2020, 2019): ANTISYMMETRY is violated when a head *is preceded by* anything that it c-commands.
- In considering outputs that violate UNIFORMITY, we must instead assign violations if a head *does not precede* elements that it asymmetrically c-commands.

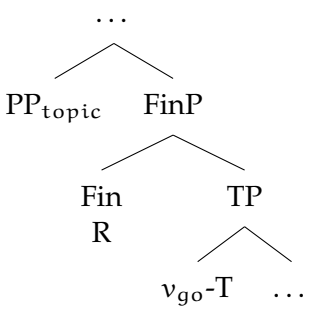
  For input: [X^0 [Y^0]]

  Violated by: $y > x$
  Satisfied by: $x > y$, **but also** $y > x > y$

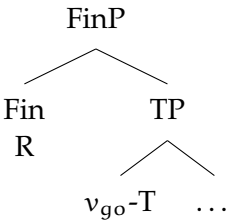
Ranking that induces doubling:

PROSODIC SUPPORT, ANTISYMMETRY ≫ UNIFORMITY

(20) [D' ar jardin] ez an.
[P DET garden] R go.1SG.
"I am going into the garden."



(21-a) **Mont a yan** d' ar jardin.
go R go.1SG P DET garden
"I am going into the garden."



(28) Linearization of (20): Topic PP occupies initial position

PP [$_{\text{FinP}}$ R [v_{go}-T ...	PR.SUPP.	ANTISYMM.	INTEG.
☞ a. $(PP)_\phi$ + R + ($(v_{go}\text{-T})_\omega$ ... $)_\phi$			
b. $(PP)_\phi$ + ($(v_{go}\text{-T})_\omega$ R ... $)_\phi$		*!	
c. $(PP)_\phi$ + $((v_{go})_\omega$ + R + $(v_{go}\text{-T})_\omega$ ... $)_\phi$			*!

(29) Linearization of (21-a): no other initial element, verb doubling the best option

[$_{\text{FinP}}$ R [v_{go}-T ... [=(21-a)]	PR.SUPP	ANTISYMM.	INTEG.
a. (R + $(v_{go}\text{-T})_\omega$ ... $)_\phi$	*!		
b. ($(v_{go}\text{-T})_\omega$ R ... $)_\phi$		*!	
☞ c. ($(v_{go})_\omega$ + R + $(v_{go}\text{-T})_\omega$ ... $)_\phi$			*

- Doubling = light verbs
- 'do' = R linearized after $\sqrt{\text{root}}$, smallest element that can generate a word to the left of *rannig*.

Again, after syntax is converted to a linear string with prosodic boundary, handed off to VI.

5 Back to predicate fronting

More widely discussed verb doubling: **predicate fronting** for topic/focus

(30) Doubling with V fronting

a. Krachi (Kandybowicz and Torrence, 2021)
Kɛ- [dɪkɛ] yɪ ɔkyɪ wʊ ɛ-dɪkɛ i-gyo.
NMLZ- cook FOC woman the PAST-cook PL-yam
"It was cooking that the woman did to yams (not, say, eating)."

b. Hebrew (Landau, 2006)
[lirkod], Gil lo yirkod ba-xayim.
[INF-dance] Gil not FUT-dance in-the-life
"As for dancing, Gil will never dance."

c. Pichi (Yakpo, 2009)
Nà [go] à dè go ò.
FOC go 1SG(subj) IMPF go PRT
"[Mind you] I'm going."

(31) Doubling with VP fronting

a. Krachi (Kandybowicz and Torrence, 2021)
Kɛ- [dɪkɛ i-gyo] yɪ ɔkyɪ wʊ ɛ-dɪkɛ.
NMLZ cook PL-yam FOC woman the PAST-cook.
"It was COOKING YAMS that the woman did (not, say, eating rice)."
"It was only cooking yams that the woman did (i.e. she did nothing else)."

b. Hebrew (Landau, 2006)
[li-knot et ha-praxim], hi kanta.
[INF-buy ACC the-flowers] she buy.PAST
"As for buying the flowers, she bought." (Hebrew: Landau (2006))

c. Pichi (Yakpo, 2009)
Nà [krach =àn] yù dè skrach.
FOC scratch =3SG(obj) 2SG IMPF scratch.
"You are actually scratching it."

Common intuition: this is **multiple copy realization**.

Something about the chain created by predicate fronting allows / requires doubling in some languages.

Alternative outputs: movement without doubling, or DO-support

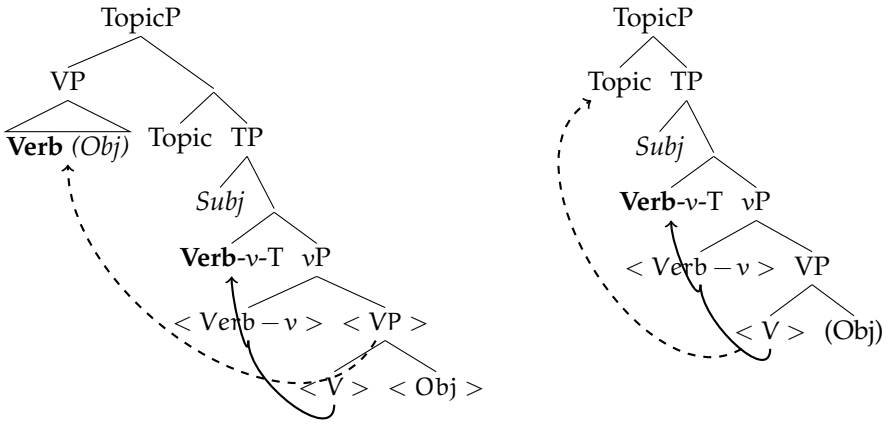
Hein (2017, 2020): an **asymmetry** between "repairs" of V and VP fronting.

- V fronting: doubling
- VP fronting: DO-support

Proposes an analysis that assumes that head movement and copy deletion are both PF operations, which can be ordered with respect to one another (Hein 2020: 118)

Alternative approach in staged realization terms

- Assumptions
  - VP fronting involves XP movement of VP to Spec,TopicP; V fronting involves head movement to Topic (whether via intermediate functional heads or as long head movement)



Proposal

- In linearization + prosodification
  - In VP fronting, if V moves out of VP, participates in **two chains**
  - In V fronting, V is treated as being in a single chain, even if it moves to v/T, because target of long head movement c-commands other links.
- In VI :
  - DO is the realization of v in the absence of a $\sqrt{\text{root}}$

Constraining Verb Doubling

- Faithfulness constraints
  - REALIZE-CHAIN: Each movement chain must have an overt realization in the output.
- Markedness constraints
  - REALIZE-HEAD: The head of a chain must be realized.

    (cf. HEAD-REAL, Müller, 2012)
  - V-OBJ: The verb must be prosodically contiguous with its direct object.

Multiple Realization in Head Movement (single chain):

- UNIFORMITY, REALIZE-HEAD≫ V-OBJ

  No verb doubling, verb can move to C.
- V-OBJ, REALIZE-HEAD≫ UNIFORMITY

  Verb doubling when verb moves to C.

6 Conclusions

The proposed architecture is one in which morphology-specific operations are limited to VI.

- See Siddiqi (2010) for discussion of conceptual advantages of limiting morphology to VI, and proposal to cover phenomena for which other mechanisms are proposed in DM with spanning.
- See also Nanosyntax Starke (2010) as a post-syntactic realizational model in which morphological operations are limited to insertion.

Morphology is not the only aspect of spell-out, however.

- Proposal here is that PF involves a series of hand-offs between aspects of realization.
- A relay from syntax to phonology, by way of linearization+prosodification and then VI.

Focusing on linearization as a component prior to realization, several extensions:

- Particle verbs in English
  - Linearization and the paradox of their constituency
  - Unexpected behaviour with agentive nominalization: *picker upper*
- Asymmetry between verb and nominal doubling
  - When **verbs** are subject to multiple copy realization: **full doubling**
  - When **nominals** are subject to multiple copy realization: **pronominal doubling**

Focusing on VI as a post-linearization process:

- Re-evaluating upwards/downwards allomorphy as inwards/outwards
  - Prefixes vs. suffixes?
- Spans vs. linearization-based merger/fusion

Thank you!

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